



Name _____

Preliminary Mathematics

Assessment Task 2 - 2013

Time Allowed - 75 minutes

Instructions: Calculators may be used in any parts of the task. For 1 Mark Questions, the correct answer is sufficient to receive full marks. For Questions worth more than 1 Mark, necessary working MUST be shown to receive full marks.

Multiple Choice	/4
Question 5	/12
Question 6	/12
Question 7	/12
Total	/40

Fill in the correct answer on the answer sheet - Questions 1 - 4 are worth 1 mark each

1. If $f(x) = -x^3 - 2x^2 - 32$ find $f(-4)$

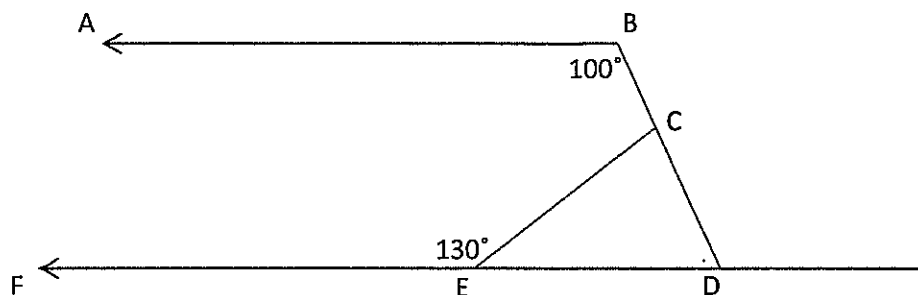
A 64

B 0

C -128

D -64

2.



In the diagram, AB is parallel to FD. The size of angle BCE is

A 100°

B 110°

C 120°

D 130°

3. $\frac{\sin(360^\circ - A)}{\sin(90^\circ - A)} =$

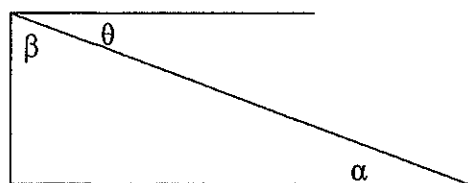
A -1

B $\tan A$

C 1

D $-\tan A$

4.



A The angle of elevation is α and the angle of depression is β

B The angle of elevation is β and the angle of depression is θ

C The angle of elevation is θ and the angle of depression is β

D The angle of elevation is α and the angle of depression is θ

Question 5 12 Marks (Begin a new sheet of paper)

Marks

a) (i) Sketch the graph of $y = \frac{1}{3-x}$

2

(ii) State its domain and range

2

b) Sketch the region defined by

$$y \geq x^2 - 4 \text{ and } y < 2x - 4$$

4

c) (i) Sketch the graph of $y = f(x)$ given

$$\begin{aligned} f(x) &= x^2 && \text{for } x > 2 \\ &= 2x - 1 && \text{for } -1 \leq x \leq 2 \\ &= -3 && \text{for } x < -1 \end{aligned}$$

3

(ii) Is this function even, odd or neither? Justify your answer.

1

Question 6 12 Marks (Begin a new sheet of paper)

Marks

a) Find the exact value of

i) $\cos 330^\circ$

1

ii) $\operatorname{cosec}(-120^\circ)$

1

b) Solve i) $\sec \theta = 2$ for $0 \leq \theta \leq 360^\circ$

1

ii) $\sin 2\theta = -\frac{\sqrt{3}}{2}$ for $0 \leq \theta \leq 360^\circ$

2

c) Simplify $(1 + \tan^2 \theta)(1 - \sin^2 \theta)$

1

d) A ship sails on a bearing of 125° from a port P, to a port Q 20 km away.

At Q, it turns and sails a further 12 km on a bearing of 215° to R.

(i) Draw a diagram with this information

1

(ii) On what bearing must it sail to return home? (nearest degree)

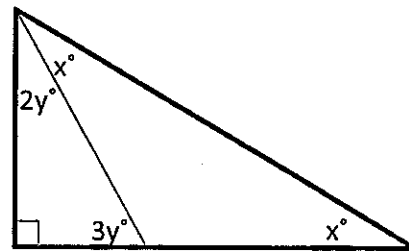
2

e) Prove $\frac{1 - \sin^2 \theta \cos^2 \theta}{\cos^2 \theta} = \tan^2 \theta + \cos^2 \theta$

3

Question 7 12 Marks (Begin a new sheet of paper)

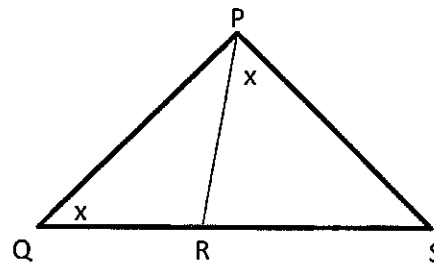
- a) Find the values of x and y



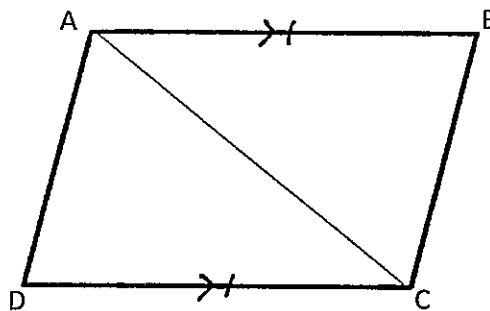
- b) In the diagram, angle $RPS = \text{angle } PQS$
 $PS = 6\text{cm}$, $QS = 9\text{cm}$ and $PQ = 7\text{cm}$.

Copy the diagram

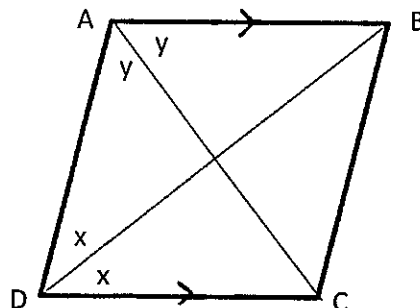
- i) Prove angle $PRS = \text{angle } QPS$
 ii) Name a pair of similar triangles
 iii) Hence find the length of PR



- c) If a quadrilateral has one pair of opposite sides both equal and parallel, then it is a parallelogram. Copy the diagram and prove this test for parallelograms, by calling your quadrilateral $ABCD$ and joining diagonal AC . (First prove $\triangle ABC \equiv \triangle CDA$)



- d) In a new quadrilateral $ABCD$, AB is parallel to DC and the bisectors of angles A and D pass through the points C and B respectively.



Copy the diagram

- i) Prove that triangle ABD is isosceles
 ii) Prove that triangle ADC is isosceles
 iii) Prove that $ABCD$ is a rhombus

END OF TEST

Solutions to 2013 Prelim Task 2

1. $64 - 32 - 32 = 0$ B

2. D

3. $-\frac{\sin A}{\cos A} = -\tan A$ D

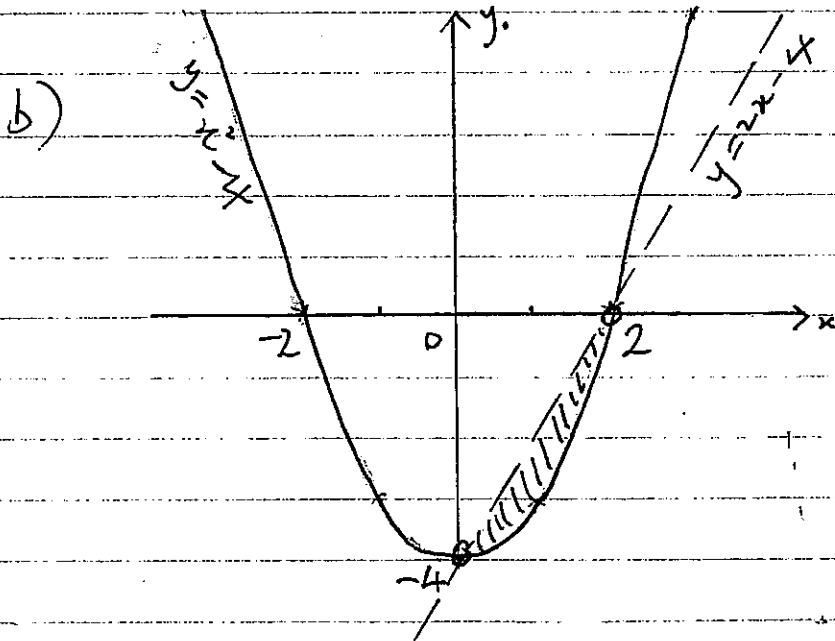
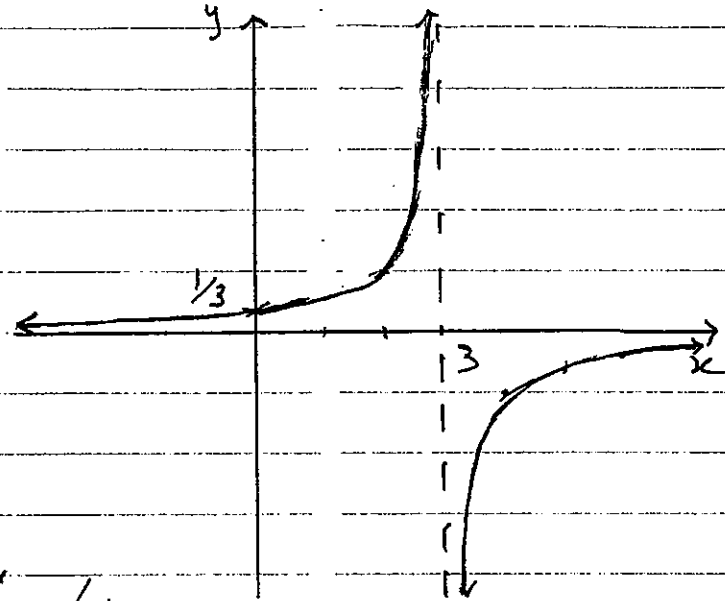
4. D

Question 5

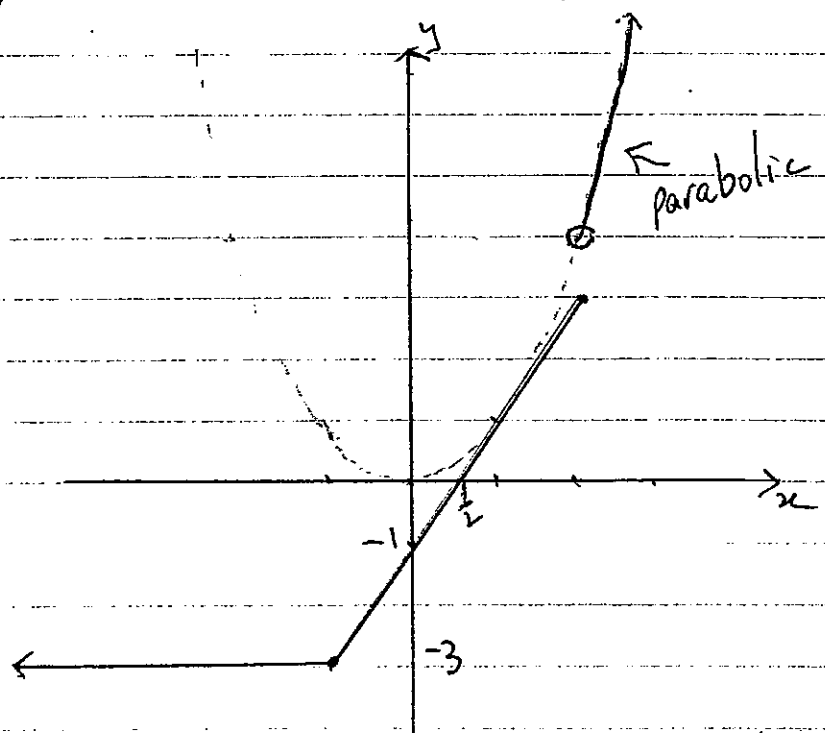
a) i) $y = \frac{1}{3-x} = \frac{-1}{x-3}$

ii) Domain: All real x
except $x=3$

Range: All real y
except $y=0$



c) i)



c) ii) Neither even
nor odd as
there is no symmetry
about y axis or
point symmetry about
origin.

Question 6

a) i) $\cos 330^\circ = + \cos 30^\circ$
 $= \frac{\sqrt{3}}{2}$

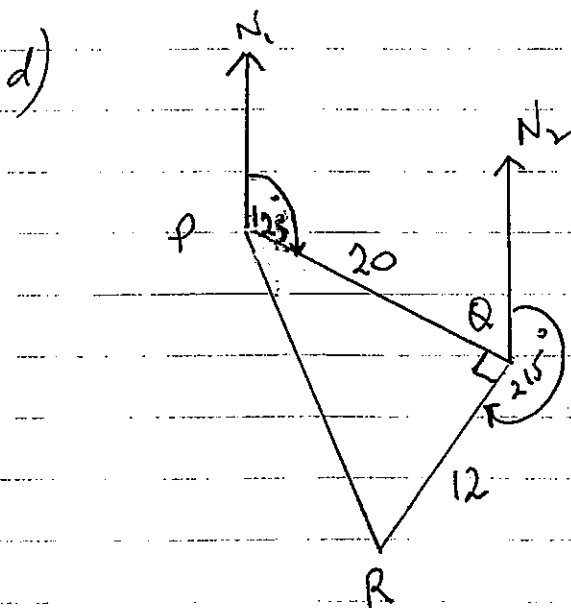
ii) $\csc(-120^\circ) = \csc 240^\circ$
 $= -\csc 60^\circ$
 $= -\frac{2}{\sqrt{3}}$

b) i) $\sec \theta = 2$
 $\cos \theta = \frac{1}{2}$
 θ lies in 1st or 4th quad
 $\theta = 60^\circ$ or 300°

ii) $\sin 2\theta = -\frac{\sqrt{3}}{2}$ $0 \leq 2\theta \leq 720^\circ$

Acute $\angle$ is 60° . 2θ lies in 2nd or 4th quad.
 $2\theta = 240^\circ$ or 300° or 600° or 660°
 $\theta = 120^\circ$ or 150° or 300° or 330°

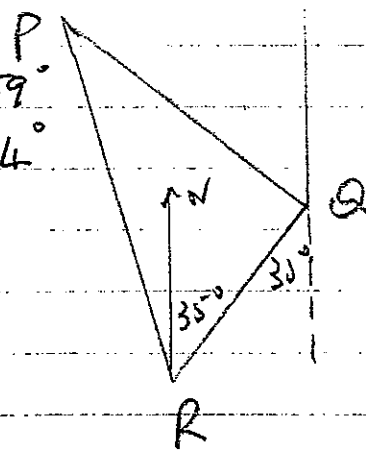
c) $(1 + \tan^2 \theta)(1 - \sin^2 \theta) = \sec^2 \theta \cdot \cos^2 \theta$
 $= 1$



$\tan R = \frac{20}{12}$
 $= 1.666...$
 $\angle R = 59^\circ$

$\angle PRQ = 59^\circ$
 $\angle PRN = 24^\circ$

$\therefore$ Bearing is
 $360 - 24^\circ$
 $= 336^\circ$



$$6c) \quad \text{LHS} = \frac{1 - \sin^2 \theta \cos^2 \theta}{\cos^2 \theta}$$

$$= \frac{1}{\cos^2 \theta} - \frac{\sin^2 \theta \cos^2 \theta}{\cos^2 \theta}$$

$$= \sec^2 \theta - \sin^2 \theta$$

$$= (1 + \tan^2 \theta) - (1 - \cos^2 \theta)$$

$$= 1 + \tan^2 \theta - 1 + \cos^2 \theta$$

$$= \tan^2 \theta + \cos^2 \theta$$

Question 7

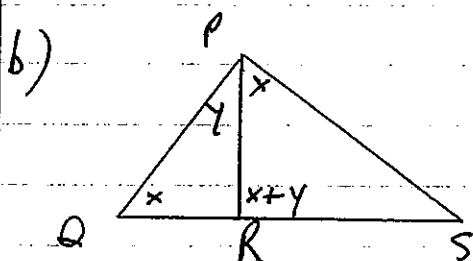
a) $2y + 3y = 90^\circ$ (angle sum of Δ)

$$y = 18$$

$2x + 2y = 90$ (angle sum of Δ)

$$2x = 54$$

$$x = 27$$



Let $\angle QPR = y$

$\therefore \angle PRS = x + y$ (exterior $\angle$ of ΔPQR)

$\angle QPS = x + y$ (sum of $\angle QPR$ + $\angle RPS$)

$\therefore \angle PRS = \angle QPS$

ii) $\therefore \Delta QPS \parallel \Delta PRS$ (2 angles test)

$$\therefore \frac{QP}{PR} = \frac{PS}{RS} = \frac{SQ}{SP}$$

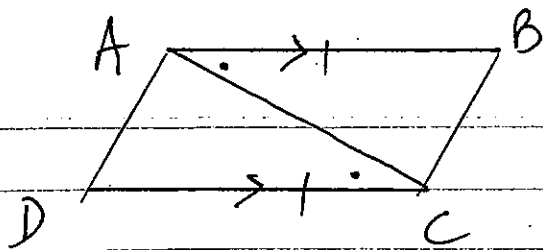
$$\frac{7}{x} = \frac{9}{6}$$

$$\frac{7}{x} = \frac{3}{2}$$

$$3x = 14$$

$$x = \frac{14}{3}$$

7c)



In Δ 's ABC, CDA

1. $AB = DC$ (given)
2. AC is common
3. $\angle BAC = \angle ACD$

(alternate $\angle$'s $AB \parallel DC$)

$\therefore \Delta ABC \equiv \Delta CDA$ (SAS test)

$\therefore \angle ACB = \angle CAD$ (matching $\angle$'s in congruent Δ 's)
But these are alternate

$\therefore BC \parallel AD$

$\therefore ABCD$ is a parallelogram.

(both pairs of opposite sides parallel)

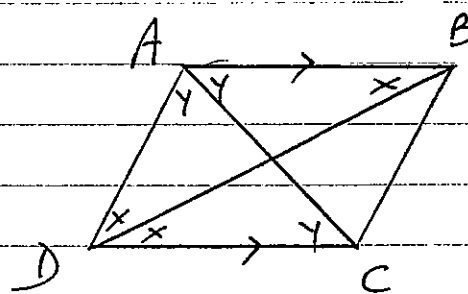
d)

i) $\angle ABD = \angle BDC =$
Alternate $\angle$'s $AB \parallel DC$

$\therefore AD = AB$

(opposite equal $\angle$'s)

$\therefore \Delta ABD$ is isosceles



ii) $\angle BAC = \angle ACD = y$ (Alternate $\angle$'s $AB \parallel DC$)

$\therefore AD = DC$ (opposite equal $\angle$'s)

$\therefore \Delta ADC$ is isosceles.

iii) $\therefore AB = AD = DC$

$\therefore$ One pair of opposite sides, AB and DC are both equal + parallel.

$\therefore ABCD$ is a parallelogram from c)

But $AB = AD$

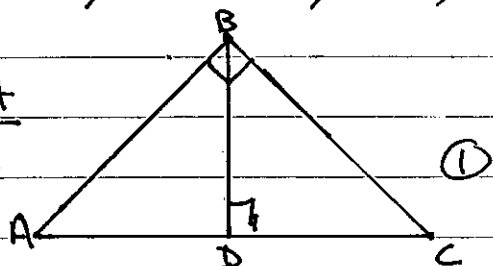
$\therefore$ One pair of adjacent sides are equal $\therefore ABCD$ is a rhombus.

Multiple Choice

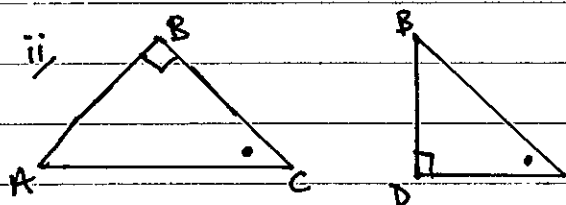
- 1, B D 2, A 3, A
4, D 5, D 6, C 7, D

PART A

a) i)

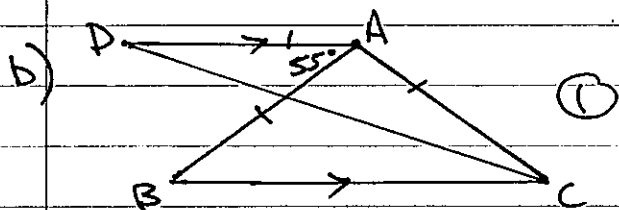


ii)



In $\triangle ABC$ and $\triangle BDC$
angle C is common (1)
angle ABC = angle BDC = 90° (given) (1)
 $\therefore \triangle ABC \equiv \triangle BDC$
(equiangular) (1)

iii $\frac{BC}{AC} = \frac{BC}{DC}$ (corresponding sides in ratio) (1)
 $BC^2 = AC \cdot DC$



ii) $\angle ABC = 55^\circ$ (alternate angle $AD \parallel BC$) (1)
 $\angle ABB = 55^\circ$ (base angles equal isos triangle) (1)

iii $\angle BAC = 180 - 2 \times 55$ (1)
 $= 70$ (angle sum of triangle)

$\therefore \angle ADC = \frac{180 - (70 + 55)}{2}$ (base angle of isos triangle ADC) (1)
 $= 27.5$

iv $\angle BCD = 55 - 27.5$ (1)
 $= 27.5^\circ$ (adjacent angle)

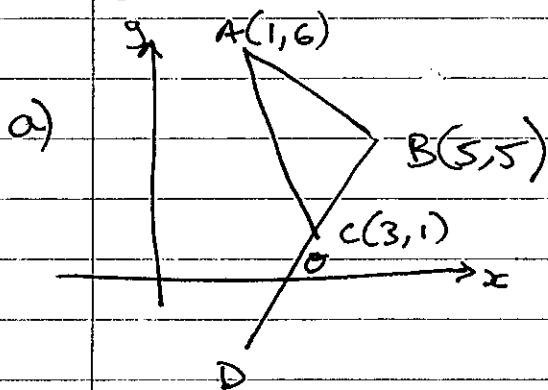
c) $\triangle AOB$ and $\triangle COD$
 $\angle AOB = \angle COD$ (vertically opposite) (1)
 $\angle BAO = \angle DCO = 90^\circ$ (given) (1)
 $BO = OD$ (given) (1)

$\therefore \triangle AOB \equiv \triangle COD$ (AAS) (1)

d) internal angle size = $\frac{(n-2) \times 180}{n}$
 $= \frac{(6-2) \times 180}{6}$ (1)
 $= 120^\circ$

$\therefore$ External angle $180 - 120^\circ$
 $= 60^\circ$ (1)

Part B.



b) i) $m = \frac{5-1}{5-3} = \frac{4}{2} = 2$ ①

c) $\tan \theta = 2$
 $\theta = \tan^{-1} 2$
 $\theta = 63^\circ$ ①

d) $y - 5 = 2(x - 5)$ ①
 $y - 5 = 2x - 10$ ①
 Eqn BD: $2x - y - 5 = 0$

e) across 2 up 4 $\therefore$ back 2 down 4
 D(3-2, 1-4) ① various methods.
 D(1, -3) ①
 $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

f) $d = \sqrt{2^2 + 4^2}$ ①
 $BC = \sqrt{20}$

$BC = 2\sqrt{5}$ ①

3

$$2x - y - 5 = 0$$

g) i) $d = \frac{|ax + by + c|}{\sqrt{a^2 + b^2}}$
 $= \frac{|2 \cdot 1 - 6 \cdot 1 - 5|}{\sqrt{2^2 + (-1)^2}}$ ①

$= \frac{|2 - 6 - 5|}{\sqrt{5}}$
 $= \frac{9}{\sqrt{5}} \text{ or } \frac{9\sqrt{5}}{5}$ ①

ii) $A = \frac{1}{2} \times 2\sqrt{5} \times \frac{9}{\sqrt{5}}$ ①
 $= 9 \text{ units}^2$ ①

iii) As $\triangle ABC$ is half $\triangle ABD$
 ratio is 1:2 ①

j) from A to B across 4 down 1
 from D to E back 4 up 1
 $\therefore E(-3, -2)$ ①

k) Area of Parallelogram = 2 $\times$ Area $\triangle ABD$ ①

$\therefore A = 2 \times 18 \text{ units}^2$
 $= 36 \text{ units}^2$ ①

Part C

(a) for $2x^2 - 3x - 4$

i/ $\alpha + \beta = \frac{-b}{a} = \frac{3}{2}$ ①

ii/ $\alpha\beta = \frac{c}{a} = -2$ ①

iii $\frac{1}{\alpha} + \frac{1}{\beta}$

$$= \frac{\alpha + \beta}{\alpha\beta}$$
 ①

$$= \frac{\frac{3}{2}}{-2}$$

$$= \frac{3}{-4}$$
 ①

iv $\alpha^2 + \beta^2$

$$= (\alpha + \beta)^2 - 2\alpha\beta$$
 ①

$$= \left(\frac{3}{2}\right)^2 + 4$$

$$= \frac{9}{4} + 4$$

$$= 6\frac{1}{4}$$
 ①

b) $4^x - 9(2^x) + 8 = 0$

let $m = 2^x$

$$\therefore 2^{2x} - 9(2^x) + 8 = 0$$

$$m^2 - 9m + 8 = 0$$
 ①

$$(m-8)(m-1) = 0$$

$$m=8 \quad m=1$$

$$\therefore 2^x = 8 \quad 2^x = 1$$
 ①

$$x=3 \quad x=0$$

3/

c) $3x^2 - 5x + 6 = A(x-2)^2 + B(x-2) + C$

$$Ax^2 - 4Ax + 4A + Bx - 2B + C$$

$$Ax^2 - (4A-B)x + 4A-2B+C$$

equate co-efficients ①

$$A=3$$

$$4A-B=5$$

$$12-B=5$$

$$B=7$$

$$4A-2B+C=6$$

$$12-14+C=6$$

$$C=8$$
 ①

$$A=3 \quad B=7 \quad C=8$$

d) $x^2 + (m-2)x + 4 = 0$

$$b^2 - 4ac = 0$$
 ①

$$(m-2)^2 - 4 \cdot 1 \cdot 4 = 0$$

$$m^2 - 4m + 4 - 16 = 0$$

$$m^2 - 4m - 12 = 0$$

$$(m-6)(m+2) = 0$$

$$m=6 \text{ or } 2$$
 ①

e) $(1+m)x^2 + 4x + m-1 > 0$

$$\therefore b^2 - 4ac > 0$$
 ①

$$4^2 - 4 \cdot (1+m)(m-1)$$

$$16 - 4m^2 + 4$$

$$-4m^2 + 20 > 0$$

$$-4(m^2 - 5)$$

$$\therefore (m+\sqrt{5})(m-\sqrt{5}) < 0$$

$$m > \sqrt{5} \quad m < -\sqrt{5}$$
 ①